
Feedback-Augmented RT-DETR

A Cross-Attention Refinement Strategy for Enhanced Small-Object Detection

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The problem: small objects in RT-DETR

RT-DETR is fast and NMS-free — but small-object AP lags.

AP_S (small)

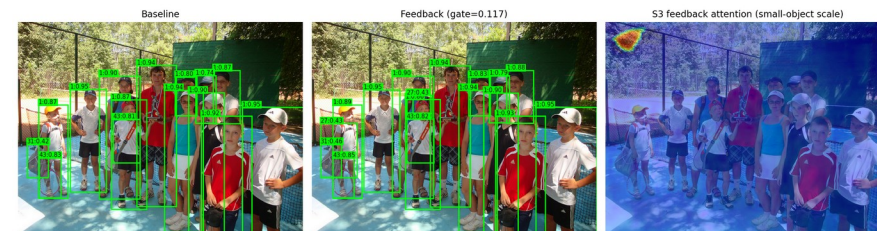
AP_L (large)

34.7

67.7

33-point gap on the same model. Why?

Encoder memory is computed once before any decoder layer fires.
It is never refined as the decoder accumulates information.
Small objects need every available cue at every layer.



The idea: feed early predictions back into memory

After decoder layer 1, refine encoder memory using its own predictions.

$$m' = \text{LN}(m + g_{\text{eff}} \cdot \text{MHA}(m, t^{(1)}, t^{(1)}))$$

Cross-attention with a learnable scalar gate g_{eff} .

$$L = \underset{\text{P2}}{160 \times 160} + \underset{\text{P3}}{80 \times 80} + \underset{\text{P4}}{40 \times 40} + \underset{\text{P5}}{20 \times 20} = 34,000$$

75% of memory tokens live at stride 4 — the natural resolution for small objects.

v1: the mechanism trained — but contributed nothing

Same-checkpoint inference ablation on COCO val2017 @ 640:

$$v1 : \Delta AP_S = 0.00$$

Toggling feedback on/off at inference made zero difference.

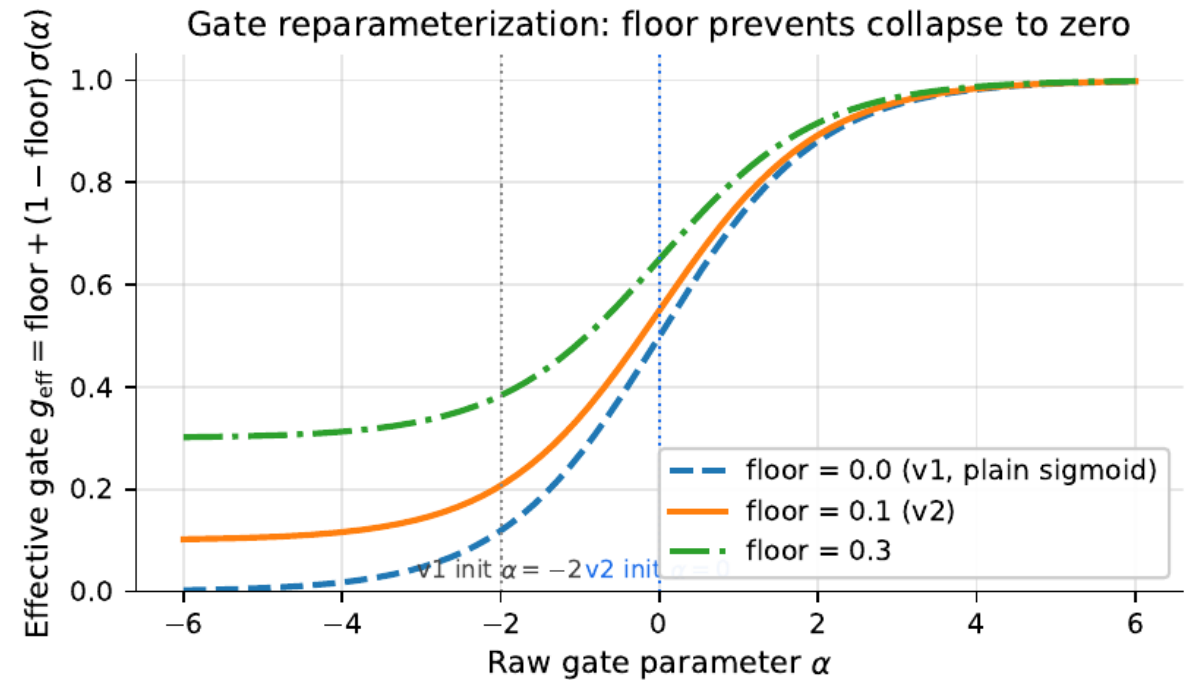
The cross-attention weights were trained, but the gate had decayed to ≈ 0.12 during training.

The feedback signal was being multiplied by $\sim$ zero before reaching memory.

Diagnosis: the gate had an escape route

Plain sigmoid gate.

- Gradient $\rightarrow 0$ as $\alpha \rightarrow -\infty$
- If the rest of the network can compensate, the optimizer pushes α arbitrarily negative at no cost.
- Nothing pulls the gate back up.



Fix 1: gate floor reparameterization

$$g_{\text{eff}} = \text{floor} + (1 - \text{floor}) \cdot \sigma(\alpha)$$

Bound the effective gate below by a floor. The optimizer can attenuate, never silence.

floor = 0.1

v2 choice

$\alpha_{\text{init}} = 0$

→ $g_{\text{eff}} = 0.55$ at start

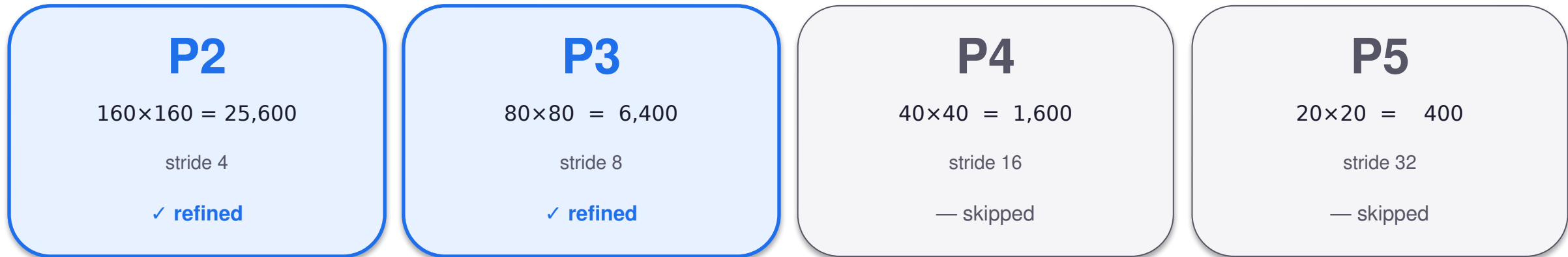
0 new params

purely a reparameterization

Fix 2: refine only where small objects live

$$\mathbf{b} = (1, 1, 0, 0) \Rightarrow \text{refine } \{P2, P3\}$$

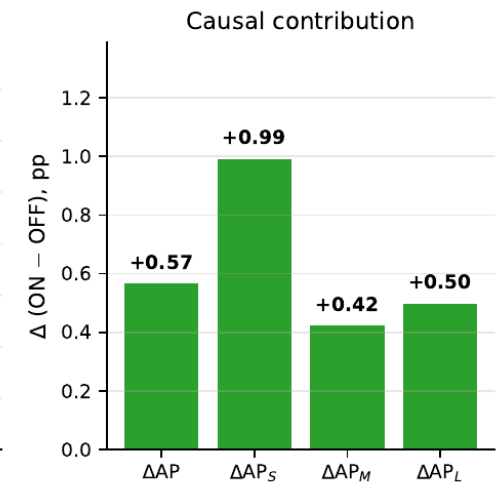
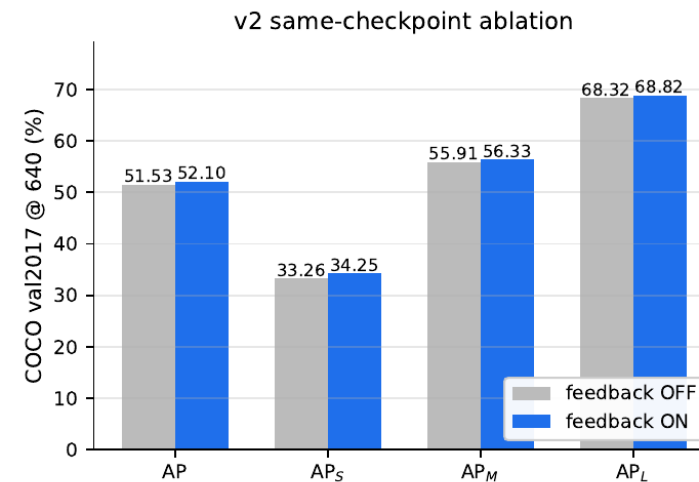
P2 (stride 4) and P3 (stride 8) carry small-object signal. S4/S5 pass through unchanged.



Result: the mechanism is now causally measurable

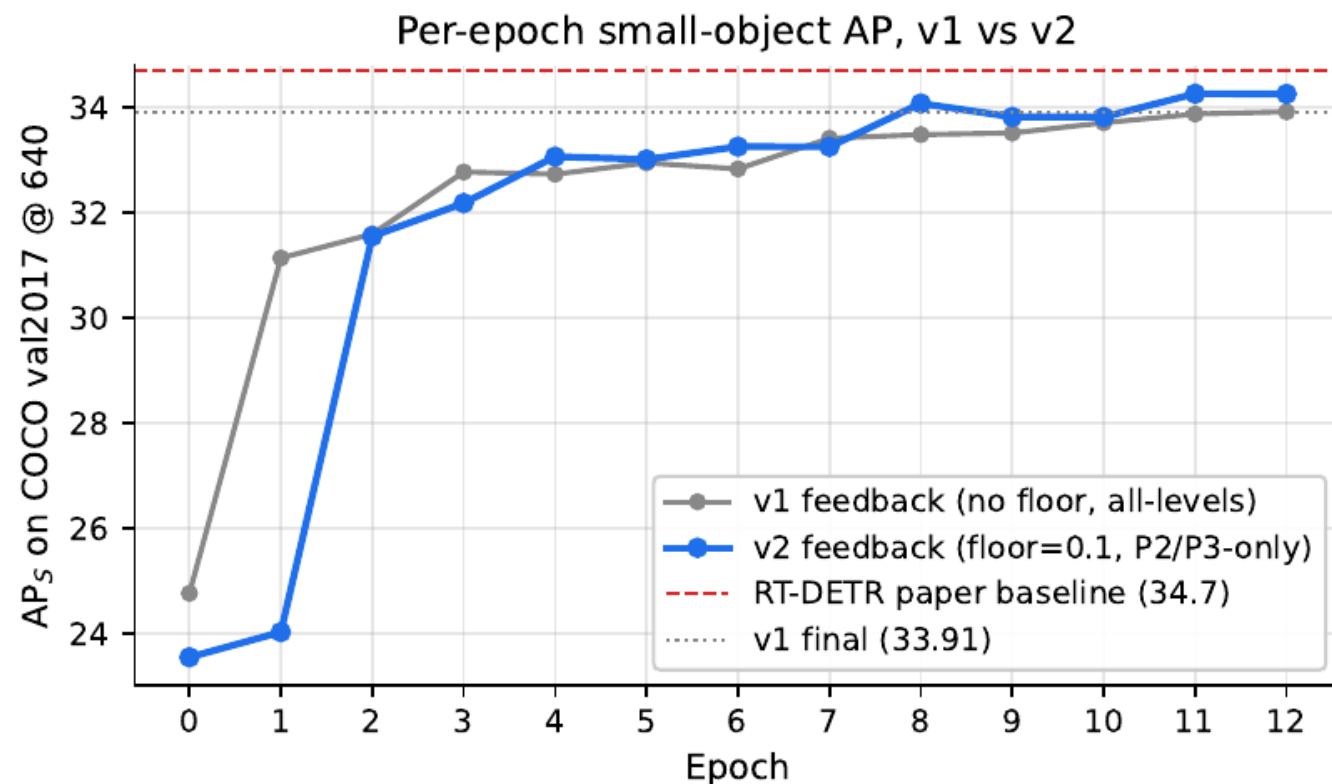
$$v2: \Delta AP_S = +0.99$$

	ON	OFF	Δ (causal)
AP	52.10	51.53	+0.57
AP_S	34.25	33.26	+0.99
AP_M	56.33	55.91	+0.42
AP_L	68.82	68.32	+0.50

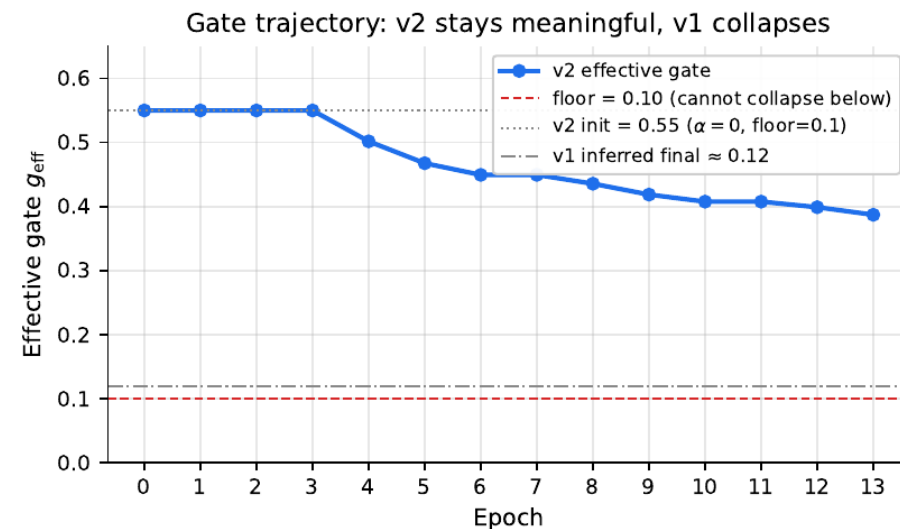


Same checkpoint, only inference-time toggle. AP_S @ 800 reaches 37.01.

v2 climbs faster and ends higher



v2 first crosses v1's eventual final 33.91 at epoch 8 — four epochs early.



Gate stays at ≈ 0.39 (well above the 0.1 floor).

Limitations and resources

Limitations

- **13 epochs** instead of the 72 of the standard 6× schedule (compute-limited finetune from public weights).
- **Single seed** — multi-seed reproducibility study is open.
- **Floor & mask not isolated** — a 2×2 ablation would resolve.
- **2× slower than baseline** — cost is from the P2 stride-4 level, not the feedback module.
- **COCO only** — aerial / surveillance datasets are obvious next steps.

Resources

 **GitHub repository**

github.com/HatemSaadallah/feedback-augmented-rtdetr

 **Full 16-page report (PDF)**

[report/main.pdf](#) · [view on GitHub](#)

 **Reproduce the ablation**

[tools/full_ablation.py](#) → `--feedback-off`

 **Course**

Computer Vision and Image Processing
Bocconi University — MSc AI

Thank you. Questions welcome.